

Network Operating Systems (NOS) Study Guide

Part I: Networking Fundamentals

Topics covered:

- Ports and Protocols
- TCP vs UDP
- Traffic Types
- Secure vs Insecure Protocols
- North-South vs East-West Traffic
- OSI Layer References
- Common Security Threats

Part II: Server Network Operating Systems

These are general-purpose operating systems designed to provide network services such as file sharing, authentication, web hosting, databases, virtualization, and cloud infrastructure.

Windows Server

Overview: Developed by Microsoft. GUI + PowerShell. Common enterprise server OS using Active Directory.

Major Elements: Server Manager, PowerShell, Event Viewer, Active Directory, Computer Management, Services Console

Common Roles: AD DS, DNS, DHCP, File Server, Print Server, Hyper-V, IIS

Management: Server Manager, RDP, Windows Admin Center, PowerShell

Advantages: Easy GUI, enterprise support

Ubuntu Server

Overview: Linux-based, free, cloud-focused, CLI by default.

Major Elements: Login prompt, Bash, APT, Systemd, Netplan

Common Roles: Web, database, Docker, Kubernetes, cloud VMs, file server

Management: SSH, Terminal, Cockpit/Webmin (optional), Ansible

Advantages: Lightweight, stable, community

Red Hat Enterprise Linux

Overview: Commercial enterprise Linux with long-term support.

Major Elements: Bash, Cockpit, DNF, SELinux, Systemd

Common Roles: Enterprise servers, databases, virtualization, cloud

Management: SSH, Cockpit, Satellite, Ansible

Advantages: Stable, secure, enterprise support

Server OS Comparison

Windows Server: Excellent GUI, PowerShell, licensed, RDP.

Ubuntu Server: Optional GUI, Bash, APT, free, SSH.

RHEL: Optional GUI, Bash, DNF, subscription, SSH.

Part III: Specialized Network Operating Systems

These run routers and switches rather than general-purpose servers.

Cisco IOS XE:

- Enterprise routers/switches
- CLI-based
- Automation support

Cisco NX-OS:

- Cisco Nexus data centers
- VXLAN/EVPN
- Feature-based architecture

Juniper Junos:

- ISP and enterprise routing
- Commit/Rollback configuration

Arista EOS:

- Linux-based
- CloudVision
- eAPI and automation

Server OS vs Network Device OS

Server OS:

- Runs applications
- Hosts databases, web servers and user services
- Supports many concurrent users

Network Device OS:

- Runs on routers and switches
- Optimized for routing and switching
- Manages packet forwarding and network infrastructure

Similarities

All support remote administration, CLI access, automation, scripting, SSH, authentication and network management.

Differences

Server operating systems provide services and applications.

Network operating systems control networking hardware and traffic.

Suggested Screenshots & Videos

Include screenshots of:

- Windows Server Manager, Active Directory and PowerShell
- Ubuntu Server terminal and Netplan
- RHEL Cockpit and terminal
- Cisco IOS XE, NX-OS, Junos and Arista EOS CLIs

Suggested introductory videos:

- Windows Server Fundamentals
- Ubuntu Server Beginner Guide
- Red Hat Enterprise Linux Basics
- Cisco IOS XE Overview
- Cisco NX-OS Introduction
- Juniper Junos Basics
- Arista EOS Overview